

WEARABLE DEVICES: Essential Inputs for Design Engineers



S.A. Srinivasa Moorthy is CEO, Andhra Pradesh Electronics and IT Agency

In the first part of this article we defined and classified wearables, and covered the designing and manufacturing phases of the life cycle of a wearable product.

In this concluding part we look at the testing and support phases, wearable system software and development phase.

Testing phase. The third important aspect is how the assembled board and product is tested. Conventionally, PCBs have test points (these are PCB pads of specific dimensions). Testers known as beds of nail testers test these assembled PCBs. These testers have a set of spring-loaded nails that are positioned against each test point. The test electronics under the control of test program tests every node by applying a stimulus and checks the response of the node and compares it with the response of a good PCB (known as gold board).

This type of testing is good when PCB sizes are not too small. However, most wearable devices use miniature PCBs and flex cables, and testing of these boards calls for special techniques. Such boards do not have space for test pads as these are crammed with components. So a frequently-used technique known as bead probe testing is used. Fig. 4 shows the conventional test pad and bead probe test pad for better understanding.

In this test method, instead of a separate test pad (typically a pad of 40mil diameter), a bead is formed (a solder bead of 20mil diameter) on the PCB trace and the tester probe probes this point and tests the PCB. This enables miniature PCBs to be tested without much difficulty. However, to implement this testing, designers have to prepare the PCB with this feature. And they need to work with the

test engineering team of the EMS partner to locate the probes for proper balancing of the PCB when testing is carried out.

Another method used, when the wearable device has only one CPU (which is normally the case), is called JTAG testing (this needs about five to seven test points, regular or bead probes). It is used to test the entire assembled PCB. This technique is popular and costs low but needs the designer's time and efforts.

One challenge in JTAG method is, if the controller IC is brand new and is being used in the product for the first time, most chip vendors may not have the required JTAG test file (known as Boundary Scan Description Language or BSDL file) ready. This can happen when the company that makes the wearable device expects a large volume and goes for a dedicated custom application specific integrated circuit, making JTAG testing difficult initially.

Finally, since most wearable devices measure very low analogue signals, these need to be calibrated once the product is assembled completely. This is the responsibility of the design team, and they need to train the EMS partner on the calibration process as well as the calibration equipment. Many a time this is missed and product shipment is delayed due to difficulty in calibration.

Use phase. This phase is where the product actually goes into the hands of the users.

One big challenge that designers face is how exhaustively they have covered the product use cases (right way of using the product) and misuse cases (wrong way of using the product). While most designers are able to cover the right use cases well, they fail to do so for the wrong use cases. In fact, testing for wrong use cases is a unique feature that medical device developers must use when developing medical devices.

Most designers think that by displaying disclaimers about wrong usage they can cover their liability. But the risk is that,

Setting devices before workout

